

Antiviral Therapy in Herpes Simplex Virus (HSV) Infection During Pregnancy and Neonatal Management

Women with primary HSV infection during pregnancy, particularly in the second and third trimesters, should receive antiviral therapy. For women at or beyond 36 weeks of gestation who are at risk for recurrent HSV infection, suppressive antiviral therapy may also be considered.

The role of HSV culture in the mother and neonate remains controversial. Current guidelines recommend performing HSV cultures at the time of delivery in women with a history of genital herpes. Women with active genital lesions at delivery and their newborns should undergo serial culturing. In neonates, specimens should be obtained from the eye, oronasopharynx, and any suspected sites of infection every 2 to 3 days during the first 4 weeks of life.

Preemptive intravenous acyclovir therapy is recommended for neonates delivered vaginally or after prolonged rupture of membranes to a mother with a clinically apparent first episode of genital herpes at delivery.

Women known to be HSV-seronegative by history and serologic testing should be counseled about the signs and symptoms of HSV infection and strategies to prevent acquisition during pregnancy. Serologic testing is particularly useful in counseling couples where the male partner has recurrent genital herpes and the pregnant woman is HSV-naïve.

Antiviral Therapy

Many HSV infections are mild and may not require specific treatment. Keeping lesions clean and dry during natural healing may be sufficient. However,

antiviral therapy is indicated for infections that are prolonged, severely symptomatic, or at risk of complications.

Acyclovir, an acyclic guanosine analogue, was the first agent shown to selectively inhibit viral replication at concentrations much lower than those affecting host cell nucleic acid synthesis. This selectivity established antiviral chemotherapy as a key medical intervention. Acyclovir has a high therapeutic index due to its preferential activation in HSV-infected cells—via viral thymidine kinase (TK)—and selective inhibition of viral DNA polymerase. It must be phosphorylated to become active, with viral TK mediating the initial monophosphorylation step.

Acyclovir inhibits HSV-1 and HSV-2 replication by 50% at concentrations of 0.1 and 0.3 µg/mL (range: 0.01–9.9 µg/mL), respectively. Toxicity occurs at concentrations >30 µg/mL. Strains requiring >3 µg/mL for inhibition are considered relatively resistant to acyclovir.

Valacyclovir, the L-valyl ester prodrug of acyclovir, provides three- to fivefold higher oral bioavailability than acyclovir and allows for less frequent dosing. **Famciclovir** is an orally absorbed prodrug of penciclovir, another guanosine analogue. After absorption, it is converted to penciclovir-triphosphate via phosphorylation. Its efficacy and safety profile are similar to acyclovir.

Topical agents include: - **Penciclovir 1% cream**, FDA-approved for treatment of herpes labialis. - **Docosanol 10% cream**, an over-the-counter FDA-approved option for recurrent herpes labialis. Docosanol is a long-chain saturated alcohol that inhibits fusion of the viral envelope with the host cell membrane. It reduces healing time by approximately 18 hours compared to placebo.

Treatment Recommendations

Antiviral regimens are tailored based on clinical presentation, immune status, and whether the infection is primary, recurrent, or being managed with suppressive therapy. Dosing recommendations vary accordingly.

For **disseminated or severe HSV infections**, including encephalitis, intravenous acyclovir is the treatment of choice: - Standard dose: 5–10 mg/kg every 8 hours. - For life-threatening infections (e.g., HSV encephalitis): some experts recommend 15–20 mg/kg every 8 hours. - **Neonatal herpes**: 20 mg/kg per dose, administered intravenously every 8 hours.

For **first episodes of genital HSV-2 infection**, oral antivirals—acyclovir, famciclovir, and valacyclovir—are effective in accelerating lesion healing, resolving symptoms, and reducing viral shedding.